# Daily Combinatorics Drill #3 — Speed Sheet

*Sheet by David Akinyele-Aje*

| Section | Topic | Questions | Target time |
|---|---|---|---|
| A | Rapid Recognition | 10 | 2 min 30 sec |
| B | Manipulation Drills | 10 | 8 min |
| C | Substitution & Structure | 8 | 10 min |
| D | Challenge | 5 | 15 min |

*New toolkit today: Blocks & Gaps · Circular Arrangements · Repeated Letters · Forced-Order Tricks.*

## Section A Rapid Recognition ($\leq$15 s each)

**1.** In how many ways can 5 people stand in a row?

Ans: $5! = 120$

**2.** How many arrangements of the letters of LEMON?

Ans: $5! = 120$

**3.** How many arrangements of the letters of BANANA?

Ans: $\frac{6!}{3!2!} = 60$

**4.** Four people around a round table (rotations the same)?

Ans: $(4-1)! = 6$

**5.** Six in a row, Ali first and Zara last?

Ans: $4! = 24$

**6.** Three identical red flags and two identical blue flags in a row?

Ans: $\binom{5}{2} = 10$

**7.** Arrangements of NOON?

Ans: $\frac{4!}{2!\,2!} = 6$

**8.** Six people around a round table (rotations the same)?

Ans: $(6-1)! = 120$

**9.** True or false: 8 people round a table = 7 people in a row?

Ans: True — both are $7! = 5040$

**10.** Arrangements of TATTY?

Ans: $\frac{5!}{3!} = 20$

## Section B Manipulation Drills ($\leq$50 s each)

**1.** Six in a row, Priya and Quinn together?

Ans: $2 \cdot 5! = 240$

**2.** Six in a row, Priya and Quinn *not* together?

Ans: $720 - 240 = 480$

**3.** Arrangements of MISSISSIPPI?

Ans: $\dfrac{11!}{4!\,4!\,2!} = 34\,650$

**4.** Five boys, three girls in a row, no two girls adjacent?

Ans: $5! \cdot (6 \cdot 5 \cdot 4) = 120 \cdot 120 = 14\,400$

**5.** Four men and four women in a row, alternating?

Ans: $2 \cdot 4! \cdot 4! = 1\,152$

**6.** Six around a round table, two particular people together?

Ans: $2 \cdot (5-1)! = 48$

**7.** Six-digit strings from $1, 1, 2, 2, 3, 3$?

Ans: $\dfrac{6!}{2!\,2!\,2!} = 90$

**8.** Arrangements of LEVEL?

Ans: $\frac{5!}{2!2!} = 30$

**9.** Seven books, trilogy in volume order (not necessarily adjacent)?

Ans: $\frac{7!}{3!} = 840$

**10.** Round-table seatings of $n$ people $= 720$. Find $n$.

Ans: $n = 7$

## Section C Substitution & Structure ($\leq$90 s each)

**1.** Eight in a row: three friends consecutive, two others (non-friends) not adjacent?

Ans: $3! \cdot 6! - 3! \cdot 2! \cdot 5! = 4\,320 - 1\,440 = 2\,880$

**2.** Same eight, no refusals, friends consecutive *and* in height order?

Ans: $6! = 720$

**3.** Arrangements of DIVIDED without all three D's together?

Ans: $\frac{7!}{3!2!} - \frac{5!}{2!} = 420 - 60 = 360$

**4.** Four men, four women alternating around a round table?

Ans: $3! \cdot 4! = 144$

**5.** Arrangements of CIRCLE with the two C's apart?

Ans: $\frac{6!}{2!} - 5! = 360 - 120 = 240$

**6.** Five-digit even numbers over 40 000 using $1, 2, 3, 4, 5$ once each?

Ans: 18

**7.** Three maths, two physics, two chemistry books, subjects together?

Ans: $3! \cdot (3! \cdot 2! \cdot 2!) = 6 \cdot 24 = 144$

**8.** Eight around a round table, two people refusing to be adjacent?

Ans: $7! - 2 \cdot 6! = 5\,040 - 1\,440 = 3\,600$

## Section D Challenge (SMC difficulty)

**1.** Seven-digit numbers from 1–7: odds increasing, evens decreasing?

Ans: $\binom{7}{4} = 35$

**2.** Ten children in a circle: Mia–Noah adjacent, Omar–Priya not?

Ans: $2 \cdot 8! - 4 \cdot 7! = 80\,640 - 20\,160 = 60\,480$

**3.** How many distinct arrangements of the letters of BANANA have exactly one of the following true: (A) the first letter is B, (B) the last letter is A, (C) no two As are adjacent? *(after TMUA Practice Paper 2 Q4)*

Ans: 23

**4.** Eight in a row, exactly two of the three sisters adjacent?

Ans: $8! - 14\,400 - 4\,320 = 21\,600$

**5.** Eight knights round a table: three feuders pairwise non-adjacent, King and Queen together?

Ans: $3! \cdot 2 \cdot (4 \cdot 3 \cdot 2) = 288$